ARUN KUMAR

Technical Lead (Manager) – Control Software

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PROFESSIONAL SUMMARY

- **8 years of progressive experience** in Managerial role as a **Technical Lead (Control Software)** in Industrial Automation, PLC programming, HMI/SCADA development, robotics integration, and MES implementation.
- Proficient in **PLC, HMI/SCADA, Robotics, Codesys, JSON, and MES** using **Structured Text (ST), Ladder Logic Programming (LLP)**, and **Function Block Diagram (FBD)**.
- Hands-on experience in debugging and troubleshooting across multiple automation domains:
  - **PLC Program Troubleshooting & Debugging** using real-time monitoring tools and forcing techniques.
  - **Firmware Debugging & Testing** in automation controllers, analysing code modules and communication protocols.
  - **Log and Error Code Analysis** for PLC, HMI, and SCADA systems to diagnose and resolve operational issues.
- Expertise in debugging fieldbus communication protocols (**Modbus, ProfiNet, TCP/IP, EtherCAT, CC-Link**) to ensure seamless data exchange between PLCs, HMIs, and sensors.
- In-depth knowledge of industrial interfaces and protocols: **TCP/IP, ProfiNet, EtherCAT, Modbus, RS-232, CC-Link.**
- Actively involved in sustaining efforts to **resolve customer-escalated and internal issues** through systematic troubleshooting.
- Hands-on experience across multiple **PLC**, **HMI**, and **robotics** platforms with strong scripting knowledge in **JSON.**
- Exceptional analytical, debugging, and problem-solving skills with a structured engineering mindset.

CORE COMPETENCIES

PLC Programming | HMI Programming | Industrial Robotics | Servo Motors | Troubleshooting & Testing | Motion Control | Networking Protocols | JSON, ADS, MQTT | SDLC | Vision Applications | Industrial Engineering | DFM | FMEA, KAIZEN | 5-Why Analysis

TECHNICAL SKILLS

PLC & Automation Platforms:

- Beckhoff TwinCAT 2/3, Mitsubishi GX-Works2 / GX-Works3 / GX-Developer, Siemens TIA Portal V16+, WPL Soft, ControlLogix, CompactLogix

Communication Protocols & Peripherals:

- TCP/IP, EtherCAT, ProfiNet, CC-Link, Modbus, RS-232, RS-485, CAN, AD, DA, PWM, PID, RTD, Timers & Counters

Programming Languages & Techniques:

- Structured Text (ST), Ladder Logic Programming (LLP), Function Block Diagram (FBD), JSON Scripting, ADS, MQTT

Tools & Technologies:

- GitHub, PLC Simulation, Real-Time Debugging, Networking & Communication Protocols, Industrial Cybersecurity, Data Management & Analytics

Quality & Manufacturing Methodologies:

- FMEA, KAIZEN, 5-Why Analysis, Design for Manufacturability (DFM), SDLC, Vision Applications

PROFESSIONAL EXPERIENCE

Tech Lead – Control Software | **VVDN Technologies** | Manesar, Haryana

September 2023 – Present

Domain: Industrial Automation – Software development for PLC, SCADA, HMI, and industrial communication systems including real-time data processing and hardware integration.

Key Responsibilities & Achievements:

- Leading a team of 8 engineers in the Control Software Department, managing project execution, task delegation, technical support, and timely delivery of automation and control solutions.
- Lead a cross-functional team in conducting foresight studies to ensure **high availability** and reliability of automation services for end users.
- Specialized in industrial automation software development ensuring efficient and reliable solutions for complex industrial processes.
- **PLC Program Development:** Designed, developed, and implemented PLC programs for **Delta**, **Beckhoff, Mitsubishi, Siemens** platforms to enable precise control and automation.
- **HMI UI Development:** Created intuitive, operator-friendly HMI designs facilitating seamless interaction between operators and automation systems.
- **Robot Integration:** Enabled seamless integration of robots with third-party PLCs, ensuring interoperability and system reliability across diverse automation setups.
- Reviewed requirements, collaborated with engineering teams on detailed design, and implemented system enhancements.

Key Projects:

Project 1: Automated DUT Handling and Validation Process (World Largest Mobile Manufacturing Company, India)

Developed and implemented an advanced automated system for handling, verifying, and processing Devices Under Test (DUTs) with high efficiency and precision. The system integrates Mac UI with a PLC-based workflow, ensuring seamless communication, real-time data processing, and high-speed automation.

- Enhanced part verification accuracy, improved robot coordination, and enabled efficient defect detection through a vision-based **Automated Optical Inspection (AOI)** system.
- Integrated **PLC**, **Mac UI**, **robots**, **sensors**, and **vision** systems for system initialization, health verification, and pre-checks to ensure operational readiness.
- Implemented safety interlocks and pre-check routines to guarantee system integrity before production starts.

Project 2: Automated PCB Connector Placement & Verification System (HP)

Designed and deployed an automated robotic system for precise connector handling and placement using vision-based inspection and PLC control.

- Implemented a vision camera to inspect trays containing 30 connectors, detect alignment, and calculate X, Y, Z, and Theta coordinates.
- Integrated coordinate data transmission from vision system to PLC via ProfiNet, with PLC subsequently transmitting corrected position data to the robot for precise placement.

Project 3: Automated Rotary Smart Meter Assembly SPM (IntelliSmart)

Engineered the IntelliSmart Rotary Base Machine — an advanced automation system for precision-driven manufacturing processes.

- Designed and implemented a **Dual-Core Rotary Mechanism** with two independent rotary cores, each equipped with **26 stations**, performing various independent tasks to enhance efficiency, accuracy, and throughput.
- Implemented synchronized dual rotary tables ensuring parallel and continuous operations, significantly reducing cycle time.

Junior Engineer – EGR & Turbochargers | **BorgWarner Emission Systems India Pvt. Ltd.** | Manesar, Haryana

April 2023 – September 2023

Responsible for the maintenance and troubleshooting of PLCs, HMIs, robots, and utility systems to ensure smooth manufacturing operations.

Senior Executive | **Mindarika Pvt. Ltd.** | Manesar, Haryana

February 2023 – April 2023

Responsible for preventive and corrective maintenance of PLC-based machines to ensure uninterrupted production.

Application Engineer – Automation Solutions | **Authentic Engineers** | Manesar, Haryana

January 2019 – February 2023

Responsible for PLC, HMI, and Robot programming for top-tier automotive manufacturers. Provided end-to-end automation solutions including design, programming, commissioning, and onsite support.

Key Clients:

Maruti Suzuki, Honda, Padmini, Hella, UNO Minda, Spark Minda, Lumax, IJL, Joyson Anand Abhishek Safety Systems.

EDUCATION

Master of Business Administration (MBA) – Information Technology

University of Delhi, School of Open Learning — **Delhi** — *Expected 2027*

Bachelor of Technology (B.Tech) – Electronics & Communications Engineering

Panipat Institute of Engineering and Technology — **Panipat** — *2013-2017*

ADDITIONAL INFORMATION

Date of Birth: 30th October, 1995

Languages Known: English & Hindi